PARMAR GK

BATCH 4.0

GEOGRAPHY

LECTURE -1

SOLAR SYSTEM

अब शुरू होगा रोला, मचेगा खेला

DIGITAL NOTES IN ENGLISH

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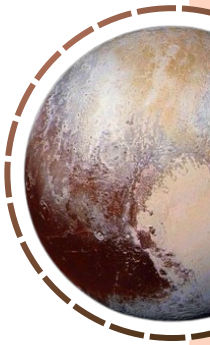
Solar System

Universe
↓
Milky Way Galaxy
↓
Solar System
↓
8 Planets



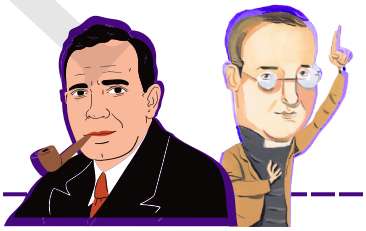
- Nearest Galaxy: **Andromede Galaxy**
- Study of Universe: **Cosmology**

Origin of Universe

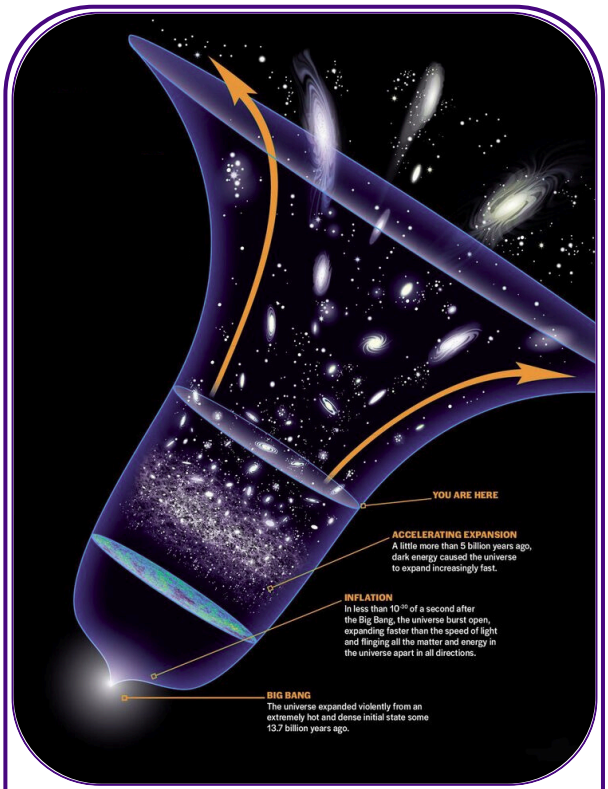
Theories given:

Big Bang Theory

- Origin of **Big Bang Theory** also known as '**Big Bang Explosion**'- Exploded **13.8 Billion years ago**.
- **George Lemaitre** (**1931**) gave the theory of **Big Bang**.



"Increase in distance of Celestial Bodies" given by George Lemaitre and Edwin Hubble.



Accelerated Expansion of Universe

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Formation of Solar System

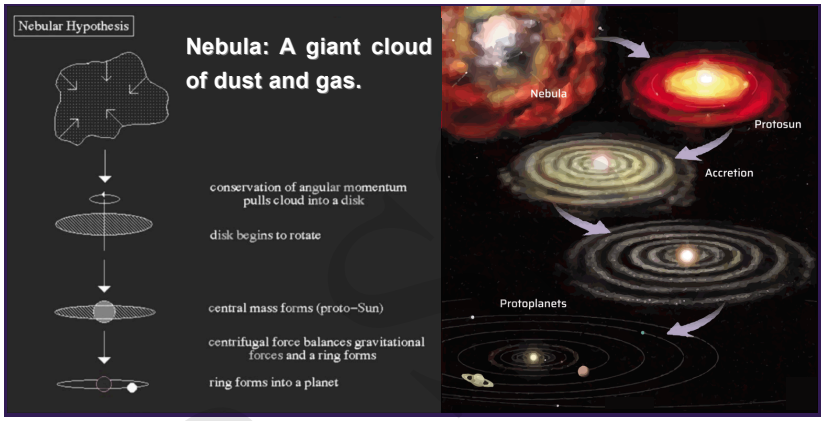
Nebula Theory

In **1755** by **Immanuel Kant.**
Remodified by **Lapse** in **1796.**

$H^+ + H^+ \longrightarrow$ **He (Nuclear Fusion)**

↓

Formation of the Sun- Primarily composed of H_2 (72%) and He.



Nebula Hypothesis

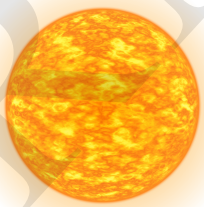
Indian Institute of Astrophysics HQ: Bangalore

Celestial Bodies

Celestial Bodies are of two types:

Luminous

Self Glowing.
Eg. **Stars (Sun)**



Non-Luminous

Not Self glowing but can reflect light through other sources.
Eg. **Moon, Asteroids, Comets, Meteor, Meteoroids.**

Stars (Luminous Bodies)

- Color depends on the temperature.
- Group of stars is called **Constellation.**

1. **Hydra:** Largest Constellation.
2. **Orion:** Brightest star Is **Rigel.**
3. **Sirius (Dog Star):** Brightest star in the Night Sky.
4. **Urja Major (Sapta Rishi):** 7 bright stars of The Big Dipper/ The Great Bear.

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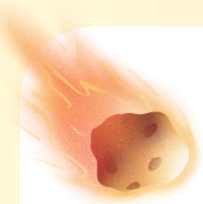
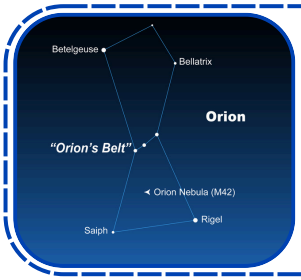


Urja Major (Sapta Rishi)

- Also known as **Great Bear** or **Big Dipper**.
- Shows **Asterism** pattern of Stars.

Hydra



Orion

Hourglass Shape

The Hunter

Cassiopiea

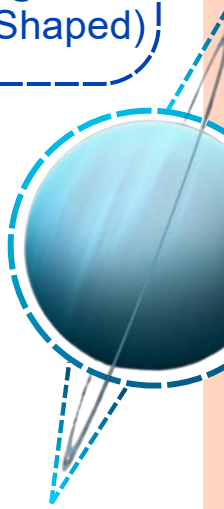
(W shaped)

Cygus

(Cross-Shaped)

Planetesimal Hypothesis

Theory about how planets were formed in Solar System by **Thomas C. Chamberlin and Forrest R. Moulton** in **1905**.



- **Light Year (LY) and Parsec measure celestial distances.**
- **1 LY = 9.63 x 10^{12} km**
- **1 Parsec = 3.26 LY**
- **After Sun Proxima Centuri is the closest star to Earth.**

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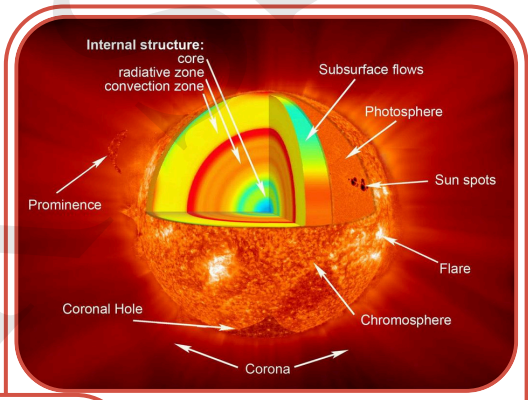
Sun (Luminous body)

- The closest star to Earth, located **150 million km (15 x 10^8)** away.
- The **only star** in and **powerhouse** of Solar System.
- Composed of **Hydrogen (73%), Helium (25%), and other metals**.
- Carries **99% mass of our Solar System**.
- Approximately **109 times mass of Earth**.
- Takes **8 Mins 30 secs** for light at speed of **3 Lakh km/sec** to reach Earth.
- Temperature at surface = **5800K/ 5600C**.
- Temperature at centre = **15.7 Million K**.
- Outer layer: **CORONA.**



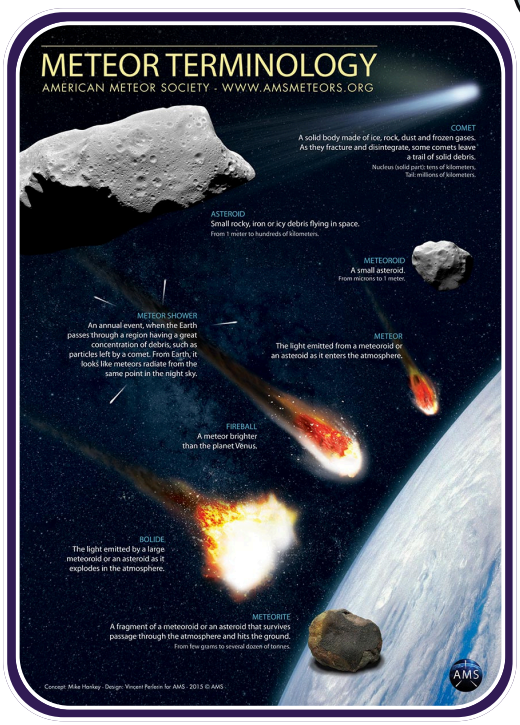
Structure of Sun

India's first Solar Mission
Aditya L1 Mission, ISRO
Lagrange Point- 5, PSLV C-57

Non-Luminous

Not Self glowing but can reflect light through other sources. ***Eg.*** **Moon, Asteroids, Comets, Meteor, Meteoroids.**

1. **Asteroid :** They Are small rocky objects that orbit the Sun.
2. **Meteor/ Meteoroids :** Enter Earth's atmosphere and burn in the Mesosphere (**Shooting star**).
3. **Comet :** Small icy dirt ball that orbit the Sun, burns upon reaching the Sun.



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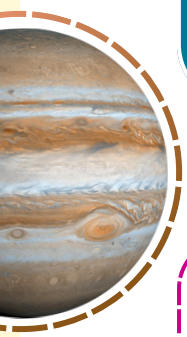
Moon (Non- Luminous Body)

- Earth's **natural satellite**.
- Radii: **1.74×10^6**.
- Time of moon's light to reach Earth: **126 secs**.
- Distance Between Earth and Moon: **3,84,000km**.
- Gravity = **1/6 Earth's Gravity**.
- **Rotation = Revolution = 27.3 days**.

(Note: Only one side of the moon is visible (**60%**) called as **Near side).**

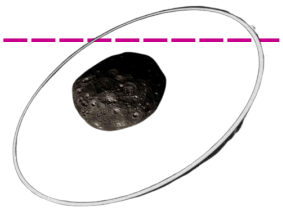
Chandrayaan 3
Lander: **Vikram**
Rover: **Pragyaan**
Point: **Shivshakti Point**



- **Rotation:** Object's spinning motion about its axis.
- **Revolution:** Object's orbital motion around another object.
- All planets rotates from West to East (anti-clockwise) except **Venus and Uranus** rotates from East to West (clockwise).

Asteroid Belt Lies between Mars and Jupiter.
Pandit Jasraj becomes the first Indian Musician to have a minor Planet named after him.

panditjasraj(300128)
(Derived from his date of birth, **28th Jan 1930**)



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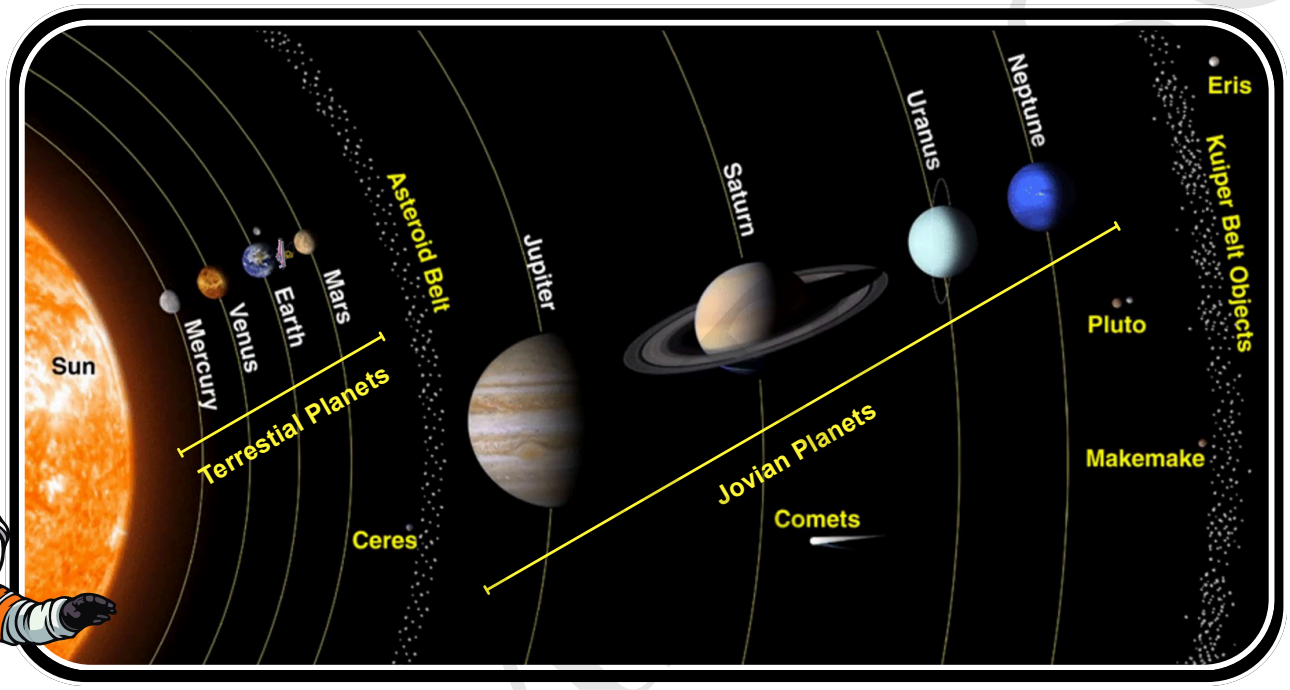
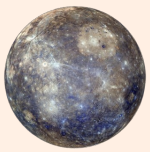
Classification of Planets

Terrestial and Jovian Planets



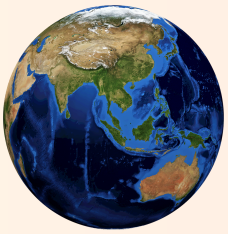
Characteristics of Terrestial Planets

- Made up of **rocky** material.
- Relatively **small**, **solid surface**.
- **Few moons** (satellites).
- **Rings are absent**.



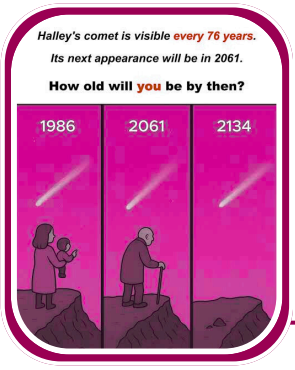
Mercury

Venus



Earth

Mars

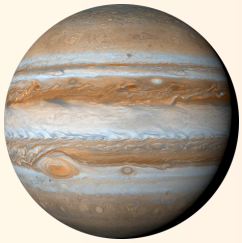


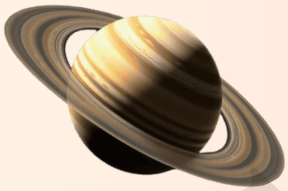
Halley's Comet Appear every **76 year.**
First observed in **1986** Next in **2061.**

Characteristics of Jovian Planets

Two Types:

Gas Gaints

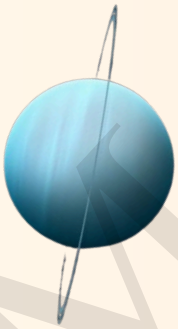


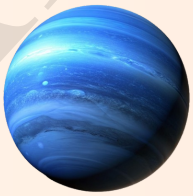


Jupiter **Saturn**

(Predominately **Helium and Hydrogen**)

Ice Gaints





Neptune **Uranus**

(Contains **rock**, **Ice**, and **mixture of Water**, **Methane** and **Ammonia**)

1st Planet: Mercury

- **Closet Planet to Sun.**
- **Smallest Planet** in Solar System.
- Diameter: **4900km**.
- **Fastest planet**, takes **88 days** to complete revolution around the Sun.
- Planet with **no satellite/ moon**.
- Planet with **no water** and **gases like Nitrogen, Hydrogen, Oxygen and Carbon Dioxide**.

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2nd Planet: Venus

- Also known as "**Earth's Twin**" due to it's mass and size.
- **Rotates Clockwise** (East to West).
- **No satellite/ Moon**.
- **Slowest Rotation: 243 days.**
- **Brightest planet in the Solar System, also known as "Evening and Morning Star".**
- **Hottest Planet in the Solar System- traps gases easily, has thick clouds of H_2SO_4 and CO_2.**

3rd Planet: Earth

- The **only planet to give support to life**.
- Also known as "**The Blue Planet**" - **70% water**.
- Has one satellite: **Moon**
- **Densest in the entire Solar System**.
- Closet Planet to Earth- **Mercury and Venus**.

4thPlanet: Mars

- Known as "**The Red Planet**"- rich in **iron oxide** (red soil).
- **Second Smallest planet** in the Solar System.
- Two natural moons: **Phobos and Deimos**.
- Largest volcano and tallest mountain of Mars: **Olympus Mons**.

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5th Planet: Jupiter

- **Largest planet** with shortest rotation- **10hrs (9hrs 56mins)**.
- Atmosphere filled with: **Hydrogen, Helium, other gases**.
- **Third Brightest** after Moon and Venus.
- Total of **95 moons** present.
- Largest satellite **Lo, Europa, Ganymede (largest among all), Callisto** (all discovered by **Galileo**).
- Has **Unclear rings** around it.
- Its is known as "**Winter Planet**".

6th Planet: Saturn

- **Second largest Planet.**
- Has bright and concentric rings made up of tiny rocks, gas, dust, ice
- It is the **least dense planet**.
- Has **146 moons** at present (the maximum).
- **Titan and Enceladus** (satellites of Saturn) shows possibilities of life on Saturn.
- Largest satellite- **Titan**
- **Huygenes (1655)** discovered **Saturn's rings.**
- **Casssini (1675)** discovered **gap between the rings** known as **"The Cassini Division".**

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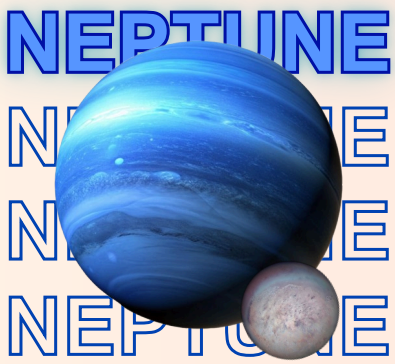
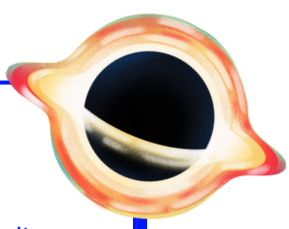
7th Planet: Uranus

- It is Greenish in Color "**Green Planet**" due to presence of **Methane (CH_4)**.
- Discovered by **William Hershel** in **1781**.
- Known as the "**Ice Giant**".
- Atmosphere has: **Hydrogen, Helium, Water, Ammonia, Methane**.
- **Rotates clockwise** (East to West) like Venus.
- **Coldest Planet**.
- It is tilted to **98°** at its axis- **Rolling/ lopsided Planet** (due to it's tilt).

8th Planet: Neptune



- Also called as the "**Ice Gaint**".
- **Farthest planet** and has **shortest revolution- 165 years**.
- Atmosphere composed of **Hydrogen, Helium**.
- **Bluish in color due to Methane**.
- Fourth largest planet and third most massive planet.
- Discovered by **Johann Galle and Urbain Le Verrier** in **1846** (only planet found by **Mathematical predictions**).
- Has **14 Satellites**, famous moon: **Triton**
- It is the **windiest planet**.



- **Supernova**: Explosion of star.
- **Pulsar**: A neutron star.
- **Black Hole**: A place in space with immense gravity.

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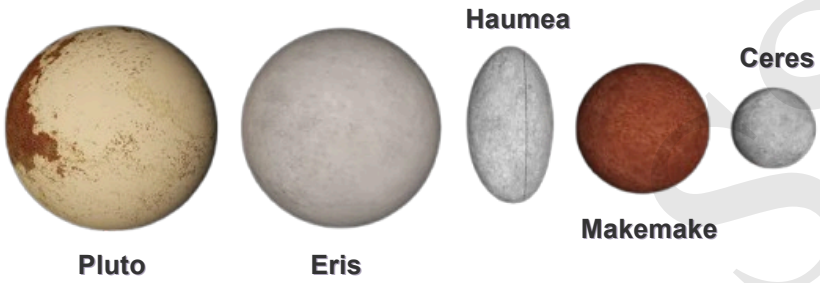
Dwarf Planets



- **Characteristics of "Dwarf Planets" are:**
  1. To be in orbit around the Sun.
  2. Have a nearly spherical shape.
  3. Should not be able to clear their orbit of debris.
- **Pluto: Largest and brightest** dwarf planet.
- **Eris: Second Largest** dwarf Planet.
- **Haumae: Fastest rotation** in Kupier belt.
- **Makemake: Third largest** and it orbits in **310 years**.
- **Ceres:** Dwarf planet present in Asteroid belt.

Pluto

- **No more a planet since 2006 by International Astronomical Union (IAU)** HQ - **Paris, France**.
- It is known as "**The Dwarf Planet**" and is a **member of Kuiper Belt**.
- Kuiper Belt is a spherical boundary outside the orbit of Neptune containing a number of **asteroids, rocks and comets**.
- Pluto's largest satellite: **Charon**.
- **1 revolution = 248 Earth years.**

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